

Blue marlin (*Makaira nigricans*) longevity estimates confirmed with bomb radiocarbon dating

Allen H. Andrews, Robert L. Humphreys, Jr., and Jeffery D. Sompaga

Abstract: The longevity of blue marlin (*Makaira nigricans*) remains unresolved. The use of fin spines and sagittal otoliths for age reading has led to unconfirmed longevity estimates near 20–30 years. Age validation has been elusive because large individuals are uncommonly caught, and a technique that can be applied to structures that provide estimates of age was absent. The use of otolith chemical signatures has been limited by sagittal otoliths that are very small — whole otolith mass of adult blue marlin rarely exceeds 10 mg for the largest fish. Recent advances in the detection limits of radiocarbon (^{14}C) with accelerator mass spectrometry — coupled with recently acquired knowledge of marine bomb ^{14}C signals spanning the tropical Pacific Ocean — have led to an opportunity to age blue marlin from small amounts of otolith material. In this study, otoliths from a recently collected 1245 lb (565 kg) female blue marlin at a measured 146-inch (371 cm) lower jaw fork length were analyzed for ^{14}C . Estimated longevity was either 12–21 or 32–44 years on the basis of bomb ^{14}C dating. Using multiple lines of evidence, it was determined that the young age scenario was most likely, with evidence for an age close to 20 years using a series of deductions in the bomb ^{14}C dating method.

Résumé : La longévité des makaires bleus (*Makaira nigricans*) est une question qui demeure non résolue. L'utilisation d'épines de nageoire et d'otolites sagittaux pour déterminer l'âge de ces poissons a mené à des estimations non confirmées de leur longévité autour de 20 ans à 30 ans. La validation des âges s'est avérée difficile parce que les grands individus sont rares et qu'il n'existait pas de technique pouvant être appliquée à des structures qui donnent des estimations de l'âge. L'utilisation de signatures chimiques d'otolites a été limitée par la très petite taille des otolites, la masse totale des otolites de mako bleu adulte dépassant rarement 10 mg pour les plus grands poissons. Des avancées récentes touchant aux limites de détection du radiocarbone (^{14}C) obtenues par spectrométrie de masse par accélérateur, jumelées à de nouvelles connaissances sur les signaux du ^{14}C dérivé d'explosions en mer à l'échelle de l'océan Pacifique tropical, offrent l'occasion d'établir l'âge de makaires bleus à partir de petites quantités de matériau d'otolite. Dans la présente étude, des otolites d'un mako bleu femelle de 565 kg (1245 lb) et de 371 cm (146 po) de longueur de la mâchoire inférieure à la fourche récemment prélevé ont été analysés pour le ^{14}C . La longévité estimée est soit de 12–21 ans ou de 32–44 ans sur la base de la datation au ^{14}C dérivé d'explosions. En invoquant divers arguments, il a été déterminé que le scénario de l'âge le plus jeune est le plus probable, plus précisément un âge d'environ 20 ans sur la base d'une série de déductions associées à la méthode de datation au ^{14}C dérivé d'explosions. [Traduit par la Rédaction]

Introduction

Blue marlin (*Makaira nigricans*) age and growth has been investigated using various techniques with which confidence in the age estimates of the largest individuals was limited. Growth structures that have held the greatest promise were the sagittal otoliths and fin spines (e.g., Hill et al. 1989; Wilson et al. 1991). In each case, the quantification of what appeared to be annual growth zones provided estimates of age and growth with longevity to nearly 20 or 30 years. Recent findings from an analysis of daily increments in blue marlin otoliths revealed there may have been early overcounting within the first year of growth (Shimose et al. 2015), but the age of adult blue marlin remains unresolved. The only temporal information on the lifespan of any adult istiophorid billfish was with tag-recapture records in which the longest at-large blue marlin was 11 years for a study period of 1954–2000 (Ortiz et al. 2003). Hence, other methods need to be employed to answer questions of longevity for these large pelagic fishes.

Validation of age estimates for tropical fishes has evolved considerably with the advent of bomb radiocarbon (^{14}C) dating (Kalish 1993). This technique is advantageous with organisms of tropical

seas that live within the well-stratified, mixed layer in which air-sea diffusion of the bomb ^{14}C signal is shared among individuals growing in this marine environment. Because the carbon source of the carbonate of otoliths and the skeletal structure of hermatypic coral is largely dissolved inorganic carbon (DIC; Druffel 1997; Campana 1999), the carbon isotope record provided within each growth structure reflects the composition of the regional environment. Hence, the use of bomb ^{14}C records in hermatypic coral to validate the age of tropical fishes using ^{14}C measurements in otolith material by direct comparison is well founded (Campana 1999, 2001). This method has been useful in life history studies at various tropical locations (e.g., Cook et al. 2009; Andrews et al. 2011, 2013, 2015, 2016a; DeMartini et al. 2018), including the pelagic environment (Kalish et al. 1996; Neilson and Campana 2008; Ishihara et al. 2017).

Recent advances in the detection limits of ^{14}C with accelerator mass spectrometry (AMS) — coupled with more comprehensive knowledge of the marine bomb ^{14}C signal in the tropical Pacific Ocean (e.g., Guilderson et al. 1998, 2000; Andrews et al. 2016a, 2016c) — have provided an opportunity to apply bomb ^{14}C dating to fishes with much smaller otoliths (Andrews et al. 2015, 2016b;

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Table 1. Sample data for the series of ^{14}C analyses performed on the blue marlin otoliths (sagitta and lapillus).

Sample (NOSAMS)	Specimen	Sample type	Mass (mg)	$\text{F}^{14}\text{C} \pm 1 \text{ SE}$	$\delta^{13}\text{C}$ (‰)	Formation period	F^{14}C age (year)	Formation scenario
M1 (OS-128256)	Grander female	Sagitta core	0.6	1.1021 ± 0.0035	-5.63	1965–1977 1988–1997	32–44 12–21	Rise Decline
M2 (OS-128264)	Same as above	Whole lapillus	0.6	1.0878 ± 0.0043	-5.08	1965–1977* 1988–1997†	12–21	Rise excluded* Decline only†
M3 (OS-129352)	Same as above	Sagitta tip	1.2	1.0695 ± 0.0026	-5.54	2006.4‡	3.3‡	Last 6–7 years (~2002–2009)
M5 (OS-130181)	60 lb juvenile	Whole sagitta	0.5	1.0943 ± 0.0034	-7.23	1995.6 ± 0.5	na	Reference — known age
M6 (OS-130180)	60 lb juvenile	Whole sagitta	0.6	1.0976 ± 0.0030	-6.88	1996.2 ± 0.5	na	Reference — known age
M7 (OS-130179)	197 lb male	Whole lapillus	1.0	1.1068 ± 0.0028	nm	1985–1996§	Uptake test	Ontogenetic ^{14}C mean¶
M8 (OS-130230)	Same as above	Whole sagitta	5.1	1.1069 ± 0.0022	-5.13	1985–1996§	Uptake test	Ontogenetic ^{14}C mean¶

Note: The grander blue marlin was collected in 2009, the other individuals in 1996. Formation period for the 60 lb juveniles is the mean date for estimated 1 year of growth. nm, not measured; na, not applicable.

*Annual integration of coral F^{14}C values from mean birth year to collection year disagrees with measured value.

†Annual integration of coral F^{14}C values from mean birth year to collection year in agreement with measured value.

‡Year from decline reference for measured F^{14}C value and corresponding age that represents mean number of formation years.

§An age of 11 years was estimated using otolith mass-to-age relation (Wilson et al. 1991).

¶Test for proportional mass growth and ^{14}C uptake between sagitta and lapillus.

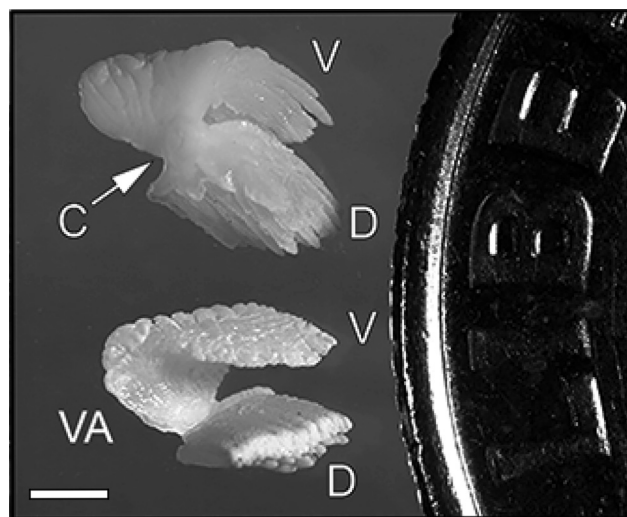
DeMartini et al. 2018) and a pelagic life history (Ishihara et al. 2017). Because the whole otolith of an adult blue marlin near maximum size typically weighs less than 10 mg (Hill et al. 1989; Wilson et al. 1991), use of core material from the first year of growth would likely be on the order of several hundred micrograms. Usually a few milligrams are necessary for AMS ^{14}C analysis with good precision (<0.5%), but instrument sensitivity has increased recently by an order of magnitude, and high precision results are attainable for carbonate sample masses near 500 μg .

A grander (≥ 1000 lb; 1 lb = 0.45 kg) female blue marlin was captured off Honolulu, Hawaii, in 2009, and the otoliths from this uncommonly caught specimen were used for ^{14}C analyses in this study. Blue marlin has been described as the most tropical of all marlins (Mather 1976; Pepperell 2010) — exhibiting a high affinity for the warmest, oxygen-rich waters (Holland et al. 1990; Block et al. 1992; Carlisle et al. 2016) — which makes bomb ^{14}C dating an appropriate method to estimate the age of blue marlin. Other smaller and informative blue marlin specimens were also analyzed — referencing coral and otolith records from across the tropical Pacific Ocean to provide temporal constraints — using a series of deductions in the bomb ^{14}C dating method that led to confirmation of age for this grander blue marlin.

Methods

An entire set of sagitta, lapillus, and asteriscus otoliths was extracted from a grander blue marlin that was captured 21 miles (34 km) south of Honolulu, Hawaii, on 1 September 2009 (Pollock 2009). This fish was approaching maximum size with a measured lower jaw fork length (LJFL) of 146 inches (3.71 m) and a total body mass of 1245 lb (565 kg). Colleagues opportunistically sampled the specimen by extracting the head, but no other hard parts or tissues were collected. The extraction of the minute otoliths from the frozen-thawed head was a challenging task owing to its enormous size (~60–75 lb = 27–34 kg with bill and gills removed). A reciprocating saw was used to sequentially trim the head into a more manageable block that contained the brain cavity and otoliths. Specifically, this process was initiated by first cutting across the top of the head (transverse section), posterior to the eye socket and straight down through the skull to a location just past the upper jaw (removed to make handling easier). This cut exposed the front of the brain cavity. Both sides of the marlin head (left and right) were then cut close to the brain cavity and excess parts were trimmed away to a block that contained the brain case. The block was then cut using a scalping method (frontal section) to expose the brain within its cavity, which was carefully removed. On either side of the brain cavity the semicircular canals with utriculus were located and extracted intact with the lapillus otoliths. The sagitta and asteriscus otoliths were removed from

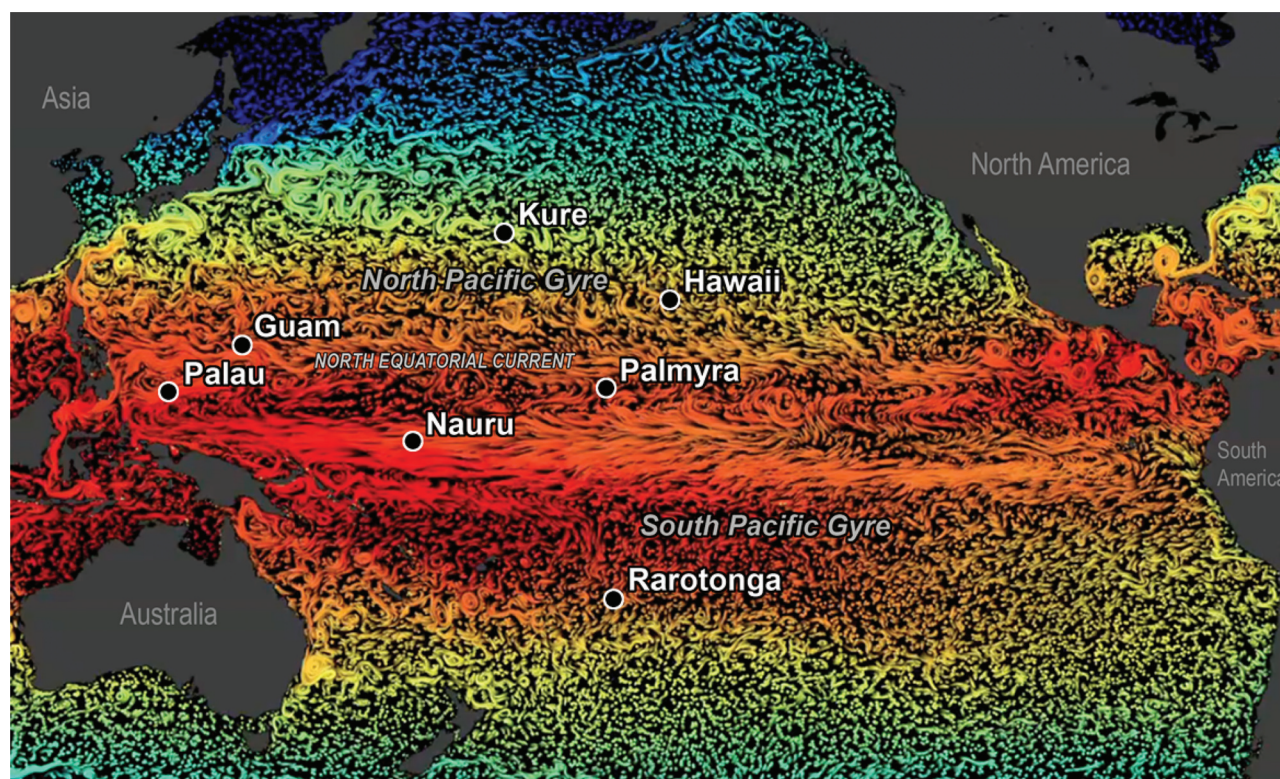
Fig. 1. Photograph of the small sagittal otoliths extracted by J.D. Sampaga from the grander blue marlin, with the edge of a US dime and 1 mm bar for scale. Dorsal (D) and ventral (V) arms are labeled with the core (C) or nuclear region and the lower ventral arm (VA) — the earliest growth sampled (sample M1).



the sacculus and lagena at a deeper location on the posterior floor of the brain cavity. All otoliths were carefully cleaned with fresh water and stored dry in a vial. Because this paper provides a call for additional otolith specimens from large blue marlin for future study, these extraction details can be used to excise a portion of the marlin skull that is easy to handle and ship (frozen) to a research facility. Otolith removal is not suggested for inexperienced persons, because they are fragile, minute in size, and difficult to locate and extract intact.

One member of each sagitta and lapillus pair was used for a ^{14}C analysis that would provide complementary results (Table 1). The sagitta mass was 8.1 mg and was analyzed in growth segments — from the core or nuclear region to the latest growth. To collect the specific otolith segments from the sagitta, each half of the u-shaped whole otolith (dorsal and ventral arms; Fig. 1) was mounted to a glass slide with Cytoseal 60 (Richard-Allan Scientific™, Thermo Fisher Scientific Inc.). Each segment was extracted manually by carefully sawing through the otolith with a curved surgical blade under a dissecting microscope (Nikon SMZ-1500). The plasticity of the mounting medium helped to confine and retain the fragile otolith portions that were isolated from the medium with clean toluene (CAS No. 108-88-3, GFS Chemicals) and weighed dry to the nearest

Fig. 2. Locations of selected tropical and subtropical Pacific Ocean coral ^{14}C records relative to general sea surface temperature (red, warmest at $\sim 30^\circ\text{C}$; yellow, cooler at $\sim 16^\circ\text{C}$) and mean current flow (line patterns in sea surface). The seven locations selected cover most of the region where blue marlin of the Pacific Ocean would reside (Su et al. 2008; Carlisle et al. 2016) and exemplify the range of expected bomb ^{14}C levels for otoliths formed while living in the mixed layer of this broad region (see Figs. 3 and 4). Map snapshot is from NASA Perpetual Oceans – Scientific Visualization Studio (Global Sea Surface Currents and Temperature).



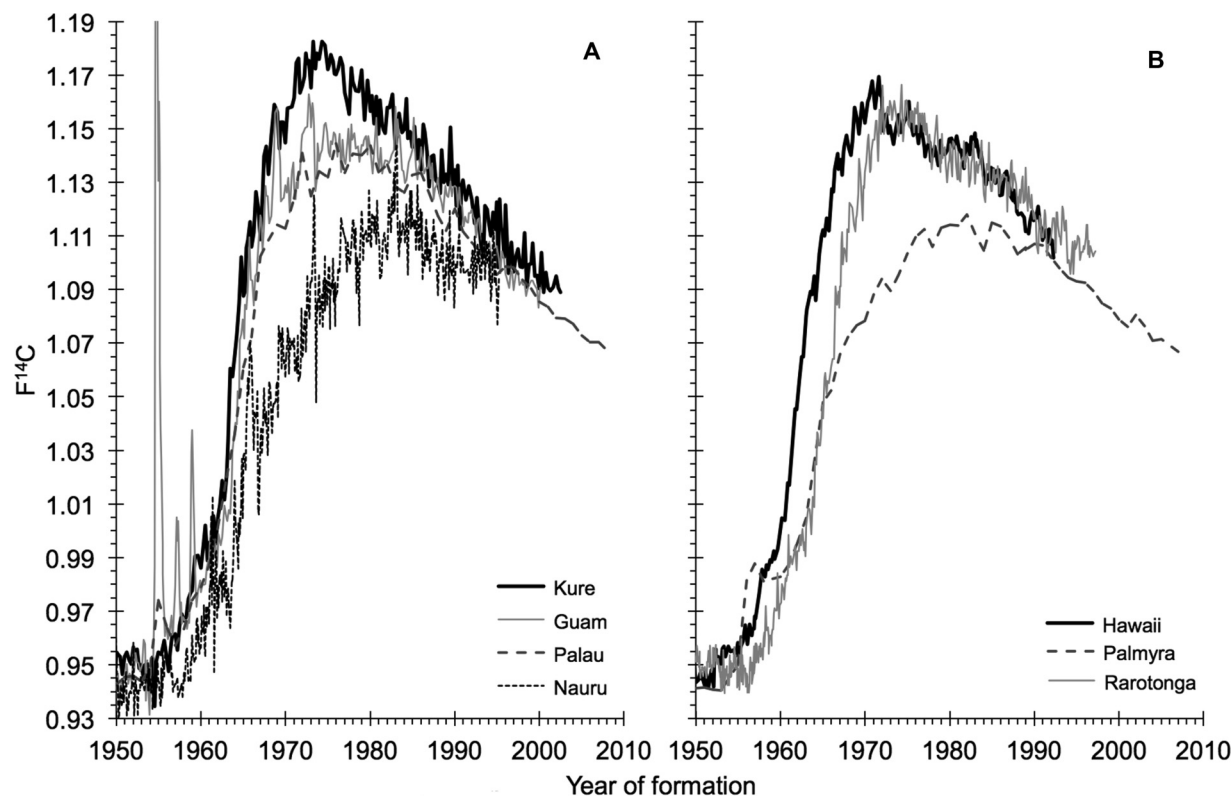
0.1 mg (Table 1). The earliest growth included the core region and the ventral arm (rostrum side) prior to its angled curvature (Fig. 1) — this portion was estimated to be most of the first year of growth (sample M1; see Shimose et al. 2015). The lapillus had a mass of ~ 0.6 mg and was analyzed whole to provide an integrated record of ^{14}C uptake through ontogeny (sample M2) — a novel tool in the refinement of bomb ^{14}C dates (DeMartini et al. 2018). The second sagittal extraction was the upper portion of the dorsal arm (antirostrum side) for a measure of ^{14}C from the latest years of life (sample M3). This piece was the outer 0.8 mm of the 2.5 mm arm length (dorsal arm mass = 3.7 mg).

To provide some verification of the methods used to deduce the age of the grander blue marlin, otoliths from smaller blue marlin collected off Oahu in 1996 were analyzed for ^{14}C . Of the 11 fish from which otoliths were extracted and available as whole specimens, two of the smallest were selected as reference material, and one adult male fish was selected to provide support for the use of the lapillus as a proxy for ^{14}C uptake through the life of the fish (Table 1). Whole sagittae were used in ^{14}C analyses of two juvenile fish (samples M5 and M6), whose ages were estimated to be ~ 1 year from the otolith mass (Wilson et al. 1991) but could have been less than ~ 6 months of accretion (60 lb (27 kg) ≈ 150 cm LJFL; Prince et al. 1991; Shimose et al. 2015). The sagitta and lapillus from the adult male (samples M7 and M8) were also analyzed whole to confirm that each provides a similar measure of environmental ^{14}C uptake by the otolith through time and to provide an additional reference point in the ^{14}C time series. The age of this fish was estimated to be 11 years old using otolith mass as a rough proxy for age (Wilson et al. 1991). This fish was 197 lb (90 kg) and may have an ca. 230 cm LJFL — near maximum size for male Pacific blue marlin (Wilson et al. 1991).

The otolith samples were submitted as carbonate to the National Ocean Sciences Accelerator Mass Spectrometry Facility (NOSAMS), Woods Hole Oceanographic Institution in Woods Hole, Massachusetts, for small sample ^{14}C analysis. Radiocarbon measurements were reported by NOSAMS as Fraction Modern — the measured deviation of the $^{14}\text{C}/^{12}\text{C}$ ratio from Modern. Modern is defined as 95% of the ^{14}C concentration of the National Bureau of Standards Oxalic Acid I standard (SRM 4990B) normalized to $\delta^{13}\text{CVPDB}$ (-19%) in 1950 AD (VPDB, Vienna Pee Dee Belemnite geological standard; Coplen 1996). Radiocarbon results were corrected for isotopic fractionation using $\delta^{13}\text{C}$ measured concurrently during AMS analysis and are reported here as $F^{14}\text{C}$ (Reimer et al. 2004). Stable isotope $\delta^{13}\text{C}$ measurements were made on a split of CO_2 taken from the CO_2 generated from acid hydrolysis (Table 1).

The ^{14}C measurements from each otolith sample provided different forms of information that were interrelated to make a conclusion about the age of the grander blue marlin. Because this specimen was from the central Pacific Ocean and because blue marlin habitat is limited to the mixed layer of tropical to subtropical waters (Su et al. 2008), it was assumed that this fish spent its life within an area of warm waters that overlap the environmental bomb ^{14}C levels recorded in coral cores across this broad region (Fig. 2). Available ^{14}C time series described from coral cores were compiled from across the region to characterize what can be considered the range of potential bomb ^{14}C uptake by the blue marlin otolith through ontogeny (Fig. 3). While other coral bomb ^{14}C records exist from waters the blue marlin could have occupied in the Pacific Ocean, all were either within or coincident over time relative to the records selected across the geographical distribution of blue marlin (Fig. 4). For the northern and southern limits in

Fig. 3. Selected tropical and subtropical Pacific Ocean coral ^{14}C records that typify the region of interest for a blue marlin that may reside at various locations through ontogeny (Fig. 2). Panels A and B were separated to reveal the patterns (differences and similarities) of the records across the broad geographic region. Panel A covers from the farthest North Pacific record at Kure Atoll (most rapid increase with greatest amplitude; Andrews et al. 2016c) through the western Pacific at Palau and Guam (intermediate peak records; Glynn et al. 2013; Andrews et al. 2016a) and into the northwestern part of the South Pacific at Nauru (phase lagged and attenuated; Guilderson et al. 1998). Panel B runs from Hawaii Island (rapid increase with intermediate amplitude) at the southern edge of the North Pacific Gyre through the central Pacific at Palmyra (attenuated due to upwelling and elevated saturation state of equatorial waters; Druffel-Rodriguez et al. 2012) and into the South Pacific Gyre at Rarotonga (rapid increase that is phase lagged and similar to Nauru; Guilderson et al. 2000). These records — coupled with all known records from across the region (Fig. 4) — provide a basis for selecting the temporal limits of the bomb ^{14}C signal taken up by the otolith of a blue marlin that lived in the region. Sharp peaks in the mid to late 1950s of panel A are associated with close-in fallout from Bikini and Eniwetok atolls that diffused into the environment (Andrews et al. 2016a).



the central Pacific Ocean, the coral bomb ^{14}C records from Kure Atoll and Rarotonga were most applicable (Guilderson et al. 2000; Andrews et al. 2016c). For the hotspots of blue marlin distribution in the central and western Pacific Ocean (Su et al. 2008; Shimose et al. 2012, 2015), coral ^{14}C records ranged from Hawaii Island and Palmyra in the central region to Guam and Palau in the west (Roark et al. 2006; Druffel-Rodriguez et al. 2012; Glynn et al. 2013; Andrews et al. 2016a). Nauru was selected to represent a more centralized position near the equator, and one that clearly represents the phase-lagged and attenuated central and western South Pacific bomb ^{14}C signal of tropical seas (Guilderson et al. 1998).

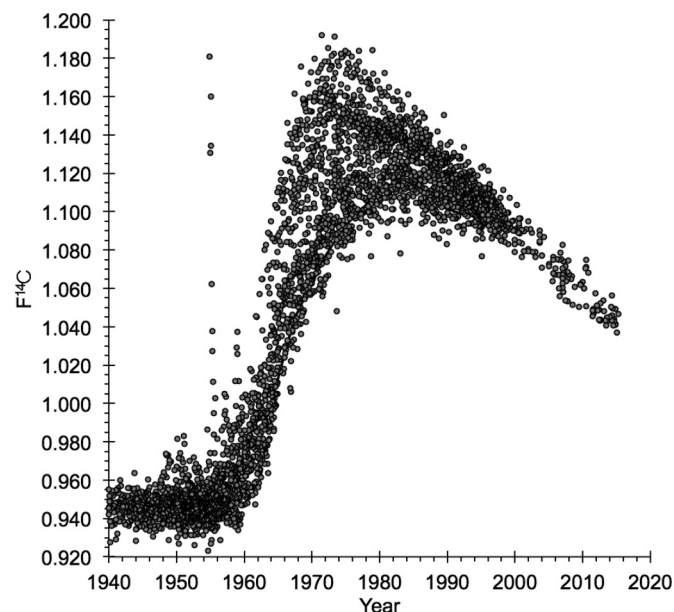
Bomb ^{14}C dating of the grander blue marlin otolith core was performed by projecting the measured ^{14}C value back through time from the collection date and across the ^{14}C reference records. Because the core represents the earliest growth, the most applicable ^{14}C records of the Pacific Ocean were based on known western and equatorial collection locations of blue marlin larvae and juveniles (Nishikawa et al. 1985; Shimose et al. 2012, 2015). The upper and lower limits selected for ^{14}C uptake were high-resolution records from the north and south central Pacific Ocean. Kure Atoll provided the most continuous record in the North Pacific — also representative of the northwestern Pacific due to exchange with the Kuroshio Current (Andrews et al. 2016c). Nauru in the central South Pacific provided the most continuous high-resolution record that exemplified the lower extreme of potential ^{14}C levels in

areas where blue marlin can reside. Each of these coral records border the extremes of ^{14}C levels and provide temporal constraints to measured levels in otolith material formed anywhere across this broad region (Figs. 2, 4).

Temporal limits for the formation of the otolith core were determined by fitting a Loess curve (spline interpolation smoothing parameter = 0.4, three-parameter polynomial, reject outliers; Sigma Plot 11.2) to the Kure Atoll and Nauru coral records (Fig. 5). Intermediate dates along the projected ^{14}C value represented the potential birth years and age scenarios. As a reference for the central western Pacific Ocean, a Guam coral record was also fitted with the Loess function — exhibiting a stronger ^{14}C signal in the areas where the early life history of blue marlin has been documented (Fig. 5). For the decline period, the regional ^{14}C records dovetail into a more constrained pattern (Fig. 4) — this portion of the record was fitted with a Loess curve that was effectively linear as a monotonic decline.

The ^{14}C measurement for the whole lapillus otolith of the grander blue marlin was used as a tool to discriminate between birth year scenarios. Lapilli grow continuously with the sagittae and there is evidence that the otolith mass growth is proportional through ontogeny (Wilson et al. 1991). There is some evidence for slightly greater accretion in the earliest lapillus growth (T. Shimose and A.H. Andrews, unpublished data), but the measured ^{14}C of the lapillus should represent a time-integrated bomb ^{14}C signal — this

Fig. 4. Composite plot of all known bomb ^{14}C data from coral and otolith carbonates from across the tropical Pacific Ocean. The F^{14}C distribution across this broad region through time provides constraints on the levels that a blue marlin of the Pacific Ocean would encounter throughout its lifespan. The upper and lower limits are typified by the coral ^{14}C records selected to describe potential uptake of ^{14}C by a blue marlin otolith (Fig. 3). This plot includes ^{14}C records ranging from the Coral Sea to the Hawaiian Archipelago and across the central and western Pacific Ocean. Excluded coral ^{14}C records were from shallow western seas and upwelled waters of the far eastern Pacific Ocean — habitat not typical of blue marlin.



assumption was tested later with a whole sagitta and lapillus otolith from the same adult male blue marlin. Assuming a simple linear mass-growth model, examination of lapillus ^{14}C uptake scenarios from the region could discriminate between young and old age scenarios (e.g., similar to a study on parrotfish otoliths; DeMartini et al. 2018). In each case, the peak bomb ^{14}C level would limit the alignment of projected ^{14}C measurements to a birth year in either (i) the pre-peak rise period (old age scenario) or (ii) the post-peak decline period (young age scenario). In general, the summation through time would be

$$\text{Total lapillus } \text{F}^{14}\text{C} = \sum_0^t \text{F}^{14}\text{C}_{(2009.7-t)}$$

where F^{14}C is the predicted Loess F^{14}C value of the chosen ^{14}C reference record at a given date $(2009.7 - t)$ at the time t during the potential lifespan of the grander blue marlin in either the young or old age scenario (2009.7 represents the collection date of 1 September 2009). This comparison of the sagitta core and whole lapillus values is only applicable when there are age scenarios separated in time (rise versus decline periods) by the bomb ^{14}C peak — a core ^{14}C value within the peak period can only provide one wide-ranging age scenario (e.g., Andrews et al. 2016b). Specifically, an integrated ^{14}C value from the rise period to recent time would be close to or above the core value due to peak period uptake, whereas the young age scenario would be strictly limited to ^{14}C values below the core value due to integrated uptake across only the decline period (depletion over time). The young and old age hypotheses were considered valid if the integrated F^{14}C value for the associated lifespan and the appropriate reference series were similar to the measured lapillus F^{14}C value (accept as not different at ≤ 2 sigma = 2 SE for $\text{F}^{14}\text{C} = \pm 0.0039$, mean error for

sagitta core and whole lapillus measurements). In addition, an otolith tip sample (latest growth, sample M3) was extracted to provide additional information that may assist with quantifying some of the assumptions about where the blue marlin resided through ontogeny and whether the young or old age scenario is accurate. This kind of sample is useful as a recent time marker and would represent an integration of ^{14}C levels from the last several years of growth prior to capture.

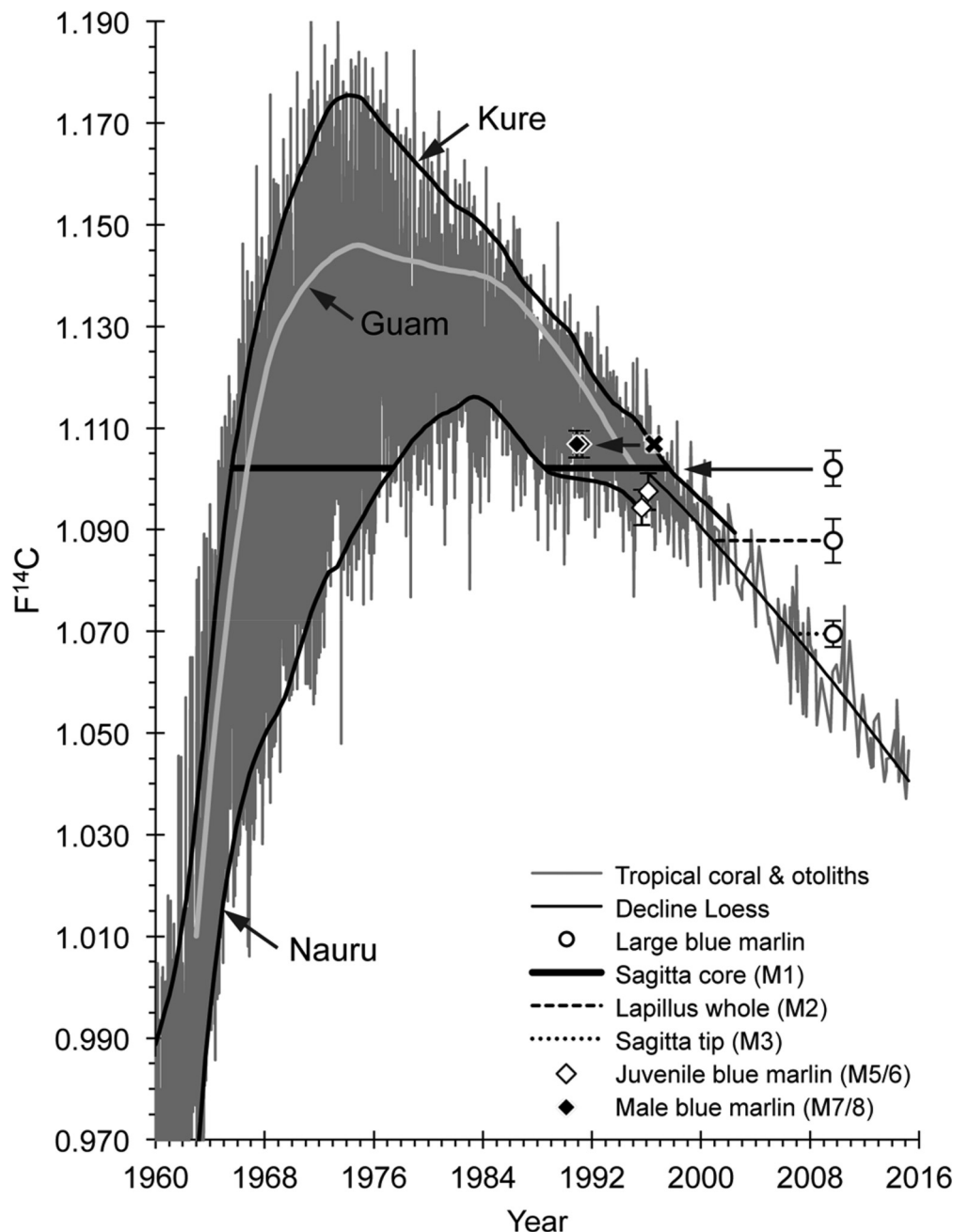
Results and discussion

Measured ^{14}C levels from otoliths of a grander blue marlin provided the first validated longevity estimates for the species using a series of deductions applied to the bomb ^{14}C dating method. Radiocarbon levels in the sagitta core material (within the first year of growth) provided evidence for either a young age or an old age scenario based on the extremes of environmental bomb ^{14}C levels (Fig. 5) — the age of this blue marlin was either 12–21 years or 32–44 years (sample M1, Table 1). Neither age scenario can be ruled out based strictly on the core ^{14}C value. However, use of the lapillus to describe ^{14}C uptake through ontogeny made the old age scenario unlikely because lifetime ^{14}C levels were significantly less than the core value (sample M2, Table 1; Fig. 5). Given a potential birth year of 1988–1997 from the young age scenario, integration over only the decline period was in close agreement with the measured lapillus value (sample M2, $\text{F}^{14}\text{C} \approx 1.09$ cf. 1.0878; within 1 sigma; Table 2), whereas potential birth years of 1965–1977 would have been equal to or more elevated than the core value ($\text{F}^{14}\text{C} \approx 1.11$ – 1.13 cf. 1.1021; up to 8 sigma greater) and 3–12 sigma greater than the lapillus because the fish would have lived through the bomb ^{14}C peak period (Table 2). Hence, because the core ^{14}C value was significantly greater than the whole lapillus ($\text{F}^{14}\text{C} = 1.1021$ cf. 1.0878; ~ 4 sigma greater), only a lifespan within the decline period can fully account for these measurement scenarios.

The age of the grander blue marlin was most likely near 20 years because the lowest-case scenario for ^{14}C uptake by the lapillus — throughout the mixed waters of this broad region — would be the unlikely scenario of an entirely Southern Hemisphere existence and an avoidance of South Pacific Gyre waters (Rarotonga coral can be used as a proxy for elevated ^{14}C levels in the gyre; Fig. 2). Integration across the attenuated signal of the southwestern Pacific Ocean (Nauru as a proxy for the lowest case ^{14}C scenario) for the old age scenario would have been greater than what was actually measured in the lapillus (Table 2). While a southern life history scenario is possible, it is unlikely because adult blue marlin gravitate to the warmest waters, regularly enter the South Pacific Gyre, and cross the equatorial Pacific Ocean (Ortiz et al. 2003; Carlisle et al. 2016).

Because this grander blue marlin was collected just south of Oahu — coupled with movement observations of adult blue marlin throughout the central tropical Pacific — a mean ^{14}C uptake scenario (between Guam and Nauru levels) was expected. In support of this notion, the otoliths from juvenile blue marlin — used as reference material collected in 1996 (samples M5 and M6) — provided evidence that the intermediate ^{14}C uptake scenario was close to what would be expected for core material from the grander blue marlin specimen (sample M1; Fig. 5). A potential complication could be a lack of applicable coral ^{14}C records in blue marlin habitat of the eastern tropical Pacific Ocean (Su et al. 2008), but coral ^{14}C records in this region are either marginal or island oriented, where upwelling of cooler waters has an impact on the bomb ^{14}C signal and does not reflect typical blue marlin habitat (coastal Panama and Galapagos Islands; Druffel 1987; Pepperell 2010). Hence, blue marlin transiting the central eastern Pacific Ocean would be constrained by the thin mixed layer (hypoxia-based habitat compression; Prince and Goodyear 2006) and would encounter ^{14}C levels similar to the range documented in the cen-

Fig. 5. Grander blue marlin otolith ^{14}C measurements plotted on the collection date (open circles) and projected back through time to the tropical Pacific coral and otolith ^{14}C records (arrow points to black bars on either side of the ^{14}C peak that represent potential formation (birth) years for the core; sample M1) that cover the geographic and bomb ^{14}C range expected to be encountered by the species. The upper and lower limits of the potential ^{14}C uptake levels (curved black lines) were a Loess curve fit to coral ^{14}C data at labeled locations (Guilderson et al. 1998; Andrews et al. 2016c). All other coral ^{14}C records (presented herein and elsewhere) were either coincident or within the abscissa–ordinate range defined by these records. The whole lapillus ^{14}C measurement provided conclusive evidence that the young age scenario was valid — only a lifespan within the decline period can fully account for these measurement scenarios (sample M2; Table 2). The sagitta tip data provided a monotonic ^{14}C decline rate confirmation (sample M3; -3.2% per year). Juvenile blue marlin provided ^{14}C reference points toward an intermediate ^{14}C uptake scenario (samples M5 and M6). Otoliths of the male blue marlin provided validation of lapillus use for integrated ^{14}C signal through ontogeny (x = collection date); data points are actually coincident, but plotted offset in time to illustrate $F^{14}\text{C}$ agreement (samples M7 and M8; Table 1). The ^{14}C records more recent than Nauru and Kure Atoll were a mix of either lower resolution coral (e.g., Palau and Palmyra) or known age otolith material (Druffel-Rodriguez et al. 2012; Glynn et al. 2013; Andrews et al. 2016a, 2016b; Ishihara et al. 2017; DeMartini et al. 2018; A.H. Andrews, unpublished data).



tral and western Pacific records (see DIC ^{14}C data from a 1974 GEOSECS transect at 124°W ; Ostlund and Stuiver 1980).

Female blue marlin in the Pacific Ocean commonly associate with the North Equatorial Current and North Equatorial Counter

Current during spawning periods, with seasonal migrations to locations farther north in the North Pacific Gyre and Kuroshio Current (Su et al. 2008; Shimose et al. 2012). These are regions of elevated bomb ^{14}C levels equal to or above the Guam coral ^{14}C

Table 2. Age hypotheses for the potential bomb ^{14}C birth years and ages with calculated uptake by the lapillus for each potential lifespan.

Age hypothesis	Reference	Birth year	Age (year)	Int- $F^{14}\text{C}$	Difference	Sigma	H_0
Old age — high	Kure Atoll	1965	44	1.1325	0.0447	11.5	Reject
Old age — low	Nauru	1977	32	1.1007	0.0129	3.3	Reject
Young age — high	Nauru	1988	21	1.0915	0.0037	1.0	Accept
Young age — low	Kure Atoll	1997	12	1.0893	0.0015	0.4	Accept

Note: Ages ranged from 12 to 21 years for the decline period and from 32 to 44 years for the rise period (Fig. 5). Birth year and age were determined by alignment of the measured core $F^{14}\text{C}$ value with either the Kure Atoll or Nauru coral $F^{14}\text{C}$ reference records in each period. The integrated $F^{14}\text{C}$ (Int- $F^{14}\text{C}$) is the predicted Loess $F^{14}\text{C}$ value of the chosen ^{14}C reference record, ranging from the hypothesized birth year to the collection year. The “Difference” value is between the calculated Int- $F^{14}\text{C}$ and the measured lapillus value ($F^{14}\text{C}$ = 1.0878). Acceptance or rejection of the age hypothesis was based on H_0 = no difference between the calculated and observed lapillus values (Accept at ≤ 2 SE for $F^{14}\text{C}$ = ± 0.0039 , mean error for sagitta core and whole lapillus measurements).

record (Figs. 3A, 5; Andrews et al. 2016a, 2016c). Given this even more elevated ^{14}C scenario and assuming this grander blue marlin followed these patterns to at least some extent, the old age scenario becomes even less likely because of the low lapillus ^{14}C levels. Hence, the young age scenario is in agreement with environmental circumstances, typical blue marlin distributions and habitat through ontogeny, and the observed otolith ^{14}C levels in each structure.

The measured ^{14}C value from the most recent sagittal otolith growth (tip of dorsal arm; sample M3, Table 1) provides additional support for the young age scenario by validating the ^{14}C decline reference records used to integrate otolith ^{14}C uptake. Because the last 0.8 mm of growth would represent several years of growth, it was expected that the ^{14}C value would be elevated from the reference record because of integration over the formation period (Fig. 5). Based on the mean age of the sample (3.3 years) determined from the slope of the declining reference record (approximately -3.2% per year), the overall age of formation for the sampled portion would be 6.6 years. This is consistent with an observed annual increment width for adult otolith growth of ~ 0.11 mm-year $^{-1}$ (calculated from increment widths in figure 6 of Hill et al. (1989)) — the period sampled would have been ~ 7.2 years of growth. A much slower growth scenario for the 0.8 mm of otolith growth would have led to a much more elevated ^{14}C value from an integration of decline levels dating back more than a decade. Hence, multiple independent lines of evidence indicate that the young age hypothesis is most likely.

The grander female blue marlin in the present study was larger than most fish that have been studied for age and growth and an age near 20 years for this grander blue marlin is consistent with other age and growth investigations. Because of what appears to be highly variable length-at-age at larger sizes in other studies, it would be difficult to reduce the margin of uncertainty for this age estimate based on fish length or otolith mass alone. Use of an otolith mass-to-age relationship indicates that the grander blue marlin might have been ~ 17 years old (Wilson et al. 1991). But in a recent study, the largest female fish approached 350 cm LJFL and was not the oldest (near 8 years), and the oldest female fish aged 20 years was not the largest (near 300 cm LJFL; Shimose et al. 2015). Another study aged the largest-ever-studied female blue marlin to 22 years at 446 cm LJFL, with the oldest fish as 27 years near 370 cm LJFL (Hill et al. 1989). However, it is important to note that early overcounting during the first year of growth — validated using daily growth increments (Prince et al. 1991; Shimose et al. 2015) — may have inflated the estimated age of individuals in other studies by several years.

Given the commonality of several regional maximum lengths and that blue marlin exceeding 350 cm LJFL are very rare (Shimose et al. 2015), the age of the grander specimen (371 cm LJFL) may represent blue marlin longevity for the fished population. In general, the findings from bomb ^{14}C dating provide support for the age and growth scenarios described using various techniques (Hill et al. 1989; Wilson et al. 1991; Shimose et al. 2015). The rare excep-

tion for lifespan potential beyond 20–30 years may be with the well-documented capture of a blue marlin from off Hawaii in 1970 that has been named “Choy’s monster” because of its unusual size (Mather 1976; Pepperell 2010). This specimen weighed 1805 lb (819 kg) at a length reported to be 434 cm LJFL (Pepperell 2002) — shorter than expected relative to its mass and apparently a fish of exceptional girth. This observation, coupled with a length of 446 cm LJFL from the largest blue marlin studied at 1650 lb (748 kg; Hill et al. 1989), provides an indication that this species may exhibit considerable growth plasticity. However, there are inconsistencies in these reported maximum lengths and masses that should be considered for what may be a more consistent trend (i.e., an ingested ~ 100 – 200 lb tuna for Choy’s monster and a shorter length of 401 cm LJFL for the 1650 lb (748 kg) specimen studied by Hill et al. (1989) — noted as “Bart’s blue” at 1656 lb (753 kg; Pepperell 2002).

The conservative age range of 12–21 years for the grander blue marlin covers all possibilities within the regional ^{14}C records for the decline period. Considering that the core sample used in the present study represented ~ 1 year of ^{14}C uptake, it is likely that the actual age was closer to 20 years because of the more likely intermediate ^{14}C uptake scenario, as evidenced by measurements from the young blue marlin reference otoliths (samples M5 and M6; Fig. 5). In other studies, in which the age of a fish was determined from either the bomb ^{14}C rise or decline period, there was usually some other information that allowed birth year refinement. The use of age reading from thin sections, otolith mass growth or fish length relations, or some combination of these factors can support the differentiation of younger versus older age scenarios (e.g., Andrews et al. 2013, 2016b). In the case of blue marlin, the rarity of otoliths — let alone large adult specimens in general — precludes an extensive comparison. Hence, the most discriminating tool available for this study — coupled with a more thorough understanding of bomb ^{14}C chronologies and the influence of oceanography — was the innovative use of the lapillus ^{14}C record to refine the temporal alignment of the sagittal otolith core measurement to corroborate the age of this grander blue marlin (Table 2). The agreement of ^{14}C uptake for the whole sagitta and whole lapillus within an adult male blue marlin validates the approach (samples M7 and M8, Table 1). In addition, the estimated age of this particular male blue marlin — assuming the otolith mass-to-age relationship is roughly valid — provides another form of support for the intermediate ^{14}C uptake scenario because ages approaching the Kure Atoll ^{14}C reference make the male blue marlin specimen too young for its size. Hence, an age closer to 20 years for the grander blue marlin is more sensible (Fig. 5).

This study highlights the utility of bomb ^{14}C dating to age tropical pelagic fishes when ocean circulation of bomb ^{14}C is understood along with the potential geographic and depth range of the species in question. As was the case with a study that validated the age of speckled hind (*Epinephelus drummondhayi*) using coral records from across the Gulf of Mexico (Andrews et al. 2013), the

geographical distribution of the study species was not necessarily limited by regional ^{14}C references. A similar circumstance was demonstrated in an age validation study of Pacific bluefin tuna (*Thunnus orientalis*) — coral ^{14}C records from across the Pacific Ocean were used to deduce age from otolith ^{14}C measurements (Ishihara et al. 2017). In the case of blue marlin, the limited range in terms of warm, oxygen-rich waters of the tropical Pacific Ocean — resulting in high mixed-layer residency (Holland et al. 1990; Block et al. 1992; Prince et al. 2010; Carlisle et al. 2016) — led to well-defined birth year scenarios from coral records that represent oceanic conditions and a validated age estimate near 20 years. While it would have been useful to have a range of specimens to better refine the age and growth of blue marlin through life — as was the case for Pacific bluefin tuna (Ishihara et al. 2017) — the rarity of otoliths sampled from large blue marlin precludes further study at this time. Hence, collection and retention of the heads from grander blue marlin, coupled with the braincase extraction technique described herein and concurrent extraction of fin spines (Kopf et al. 2010), are encouraged to make these structures available for further study.

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